

Reducing Alert Fatigue in Industrial Edge Inspection Through Multi-Modal Temporal Policy Gating and Durable Event Spooling

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Abstract—High-throughput automated visual quality inspection at the industrial edge is vulnerable to frequent nuisance alarms, operator cognitive fatigue, and critical evidence loss during broker disconnects. We present an edge inspection runtime architecture combining patch-level visual anomaly scores, time-aligned simulated physical telemetry within controlled workloads, and finite-state temporal policy gating. Our multi-modal policy framework executes optical clarity verification, dual exponential moving-average smoothing, k -of- N sliding-window confirmation, operational-state suppression, cross-modal divergence triage, and an incident refractory cooldown state machine. An embedded SQLite write-ahead-log spooler retained all 120 generated event records with zero observed missing identifiers during the evaluated 30-second simulated broker-disconnection test. Queue occupancy peaked at 120 records against the configured 50,000-record capacity under SQLite WAL `synchronous=NORMAL` persistence and MQTT QoS 1 at-least-once delivery. Evaluated across 8 policy modes, 6 standardized synthetic industrial workloads, and offline evaluation of NASA-informed degradation proxies with percentile bootstrap confidence intervals ($B = 2,000$), the runtime records observed 100.0% false-alarm suppression in the evaluated synthetic transient-glitch workload, 93.3% nuisance-alert reduction with a bounded 3.0-frame delay under sustained-defect simulation, and an end-to-end execution latency of 10.7 ms with 0.0% deadline misses under injected forward-pass latency against the 33.333 ms cycle target.

Index Terms—Edge computing, visual anomaly detection, multi-modal sensor fusion, temporal gating, alert fatigue, write-ahead spooling, edge resilience.

I. INTRODUCTION

Industrial visual inspection research increasingly considers deep feature extractors, including PatchCore [1] and PaDiM [2], for deployment-oriented anomaly detection on resource-constrained edge accelerators [3]. While achieving strong pixel AU-PRO and image AUROC under laboratory conditions [4], their deployment viability can be constrained by three operational vulnerabilities: (1) **Alert Fatigue and Cognitive Overload**: Single-frame thresholding induces bursts of nuisance alerts from reflections, dust, and non-defect vibrations. Modeling operator review capacity at $\mu = 60$ reviews/hr [5], excessive arrival rates destabilize triage queues ($\rho \geq 1.0$), risking that genuine defect incidents are overlooked. (2) **Edge Network Volatility and Evidence Loss**: Industrial plants experience

frequent network transients and broker restarts. Unbuffered edge clients risk discarding locally generated evidence during broker partitions unless persisted in durable local storage. (3) **Cycle-Time Constraints**: Defect arbitration and multi-modal sensory alignment must complete within the production line cycle time under the expected operating workload (≤ 33.333 ms) to prevent uncoordinated line stoppages.

Prior industrial anomaly-detection literature primarily emphasizes visual ranking on static benchmarks [1], [2], [4], whereas edge-AI studies prioritize model compression [3]. Alarm standards provide operational guidelines for operator workload [5], and transport protocols address broker messaging semantics [6], [7]. This work does not propose a new deep anomaly backbone or factory-floor validation; rather, it conducts an operational systems evaluation of how multi-modal temporal gating, divergence triage, queue constraints, and local event spooling interact under controlled inspection workloads. All empirical comparisons are internal ablations against a raw single-frame threshold baseline; we do not benchmark against external alarm-rationalization or sensor-fusion schemes in this initial study. Alarm management standards, including ISA-18.2, mandate that nuisance alarm floods be suppressed to prevent operator desensitization and safety degradation. Consequently, industrial edge architectures require temporal arbitration, cross-modal divergence triage, and fault-tolerant local buffering before critical dispatch.

To resolve these challenges, this paper introduces an edge inspection runtime unifying multi-modal temporal policy gating with local write-ahead event spooling. Our contributions are fourfold: (1) we design an edge runtime architecture combining PatchCore-style visual anomaly scores with time-aligned simulated physical telemetry; (2) we formulate a multi-stage temporal gating policy incorporating dual moving averages, k -of- N sliding confirmation, cross-modal divergence handling, and a refractory cooldown state machine; (3) we deploy an embedded SQLite write-ahead-log (WAL) spooler delivering zero observed missing event records during evaluated partition tests within the configured spool capacity; and (4) we evaluate eight policy variants across standardized industrial workloads, synthetic corruptions, and offline evaluation of

NASA-informed bearing and turbofan degradation proxies with bootstrap confidence intervals.

II. SYSTEM ARCHITECTURE AND FORMULATION

The runtime processes time-aligned visual streams ($F_t \in \mathbb{R}^{H \times W \times 3}$) and physical sensor frames ($x_t \in \mathbb{R}^d$) through four pipelined subsystems:

A. Visual Embedding and Physical Telemetry Extraction

Incoming video frames pass through an optical clarity check computing Laplacian blur variance σ_{blur}^2 and luminance range; frames with $\sigma_{\text{blur}}^2 < \tau_{\text{blur}}$ trigger optical degradation flags. The runtime accepts PatchCore-style visual anomaly scores $v_t \in [0, 1]$ from the configured inference path. Benchmark timing evaluates streaming scores with controlled forward-pass latency injection (8.20 ± 0.25 ms), parameterized for the 224×224 path to isolate policy and transport overheads without running live deep feature extraction. Concurrently, tri-axial accelerometer and thermocouple readings are mapped via sliding statistics into composite physical score $s_t \in [0, 1]$.

B. Multi-Modal Temporal Gating Policy

Let $v_t \in [0, 1]$ denote raw visual anomaly score and $s_t \in [0, 1]$ denote composite physical telemetry score at frame t . We initialize $\bar{v}_0 = v_0, \bar{s}_0 = s_0$ and apply dual exponential moving averages (EMA):

$$\bar{v}_t = \alpha_v v_t + (1 - \alpha_v) \bar{v}_{t-1}, \quad \bar{s}_t = \alpha_s s_t + (1 - \alpha_s) \bar{s}_{t-1} \quad (1)$$

where $\alpha_v = 0.35$ and $\alpha_s = 0.25$.

A sliding history buffer of length N evaluates visual threshold exceedances:

$$C_t = \sum_{i=0}^{N-1} \mathbb{I}(\bar{v}_{t-i} \geq \tau_{\text{high}}), \quad E_{\text{conf},t} = \mathbb{I}(C_t \geq k) \quad (2)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function, $k = 4$, $N = 10$, and $\tau_{\text{high}} = 0.80$. Unfilled positions prior to N frames are treated as non-exceedances.

Cross-modal divergence between vision and physical telemetry is evaluated on the smoothed states: $\Delta_t = |\bar{v}_t - \bar{s}_t|$. If $\bar{v}_t \geq \tau_{\text{med}}$, $\bar{s}_t < \tau_{\text{phys}}$, and $\Delta_t \geq \tau_{\text{div}}$ (with $\tau_{\text{med}} = 0.50$, $\tau_{\text{phys}} = 0.65$, and $\tau_{\text{div}} = 0.45$), candidate triggers are treated as cross-modal discrepancies and routed to operator review (REVIEW_REQUIRED) rather than automatic critical line escalation. When physical telemetry channels are unavailable or degraded, elevated visual risk is routed to REVIEW_REQUIRED unless independently corroborated.

Inspection events are processed through a seven-stage priority cascade: (1) **Optical Health:** If $\sigma_{\text{blur}}^2 < \tau_{\text{blur}}$, disable vision-only escalation and emit degradation status. (2) **State Update:** Compute smoothed EMA states ($\bar{v}_t, \bar{s}_t$). (3) **Machine Gating:** Suppress alerts during non-operational states (IDLE/MAINTENANCE). (4) **Refractory Cooldown:** If incident timer T_{cool} is active, aggregate alarms under the existing incident ID. (5) **Multi-Modal Confirmation:** Escalate to critical defect if visual persistence ($E_{\text{conf},t} = 1$) is corroborated by telemetry ($\bar{s}_t \geq \tau_{\text{phys}}$). (6) **Cross-Modal Triage:** Route to REVIEW_REQUIRED when $\bar{v}_t \geq \tau_{\text{med}}$,

$\bar{s}_t < \tau_{\text{phys}}$, and $\Delta_t \geq \tau_{\text{div}}$. (7) **Durable Dispatch:** Emit an auditable event to the local SQLite WAL spooler.

Parameterization and Failure Bounds: Constants ($\tau_{\text{high}} = 0.80$, $k = 4$, $N = 10$, $\tau_{\text{div}} = 0.45$, $T_{\text{cool}} = 15$) were calibrated on representative baselines; formal sensitivity sweeps remain future work. While cooldown suppresses redundant re-alerts for sustained defects, triggers within T_{cool} frames aggregate under the active incident ID, trading multi-incident discrimination for operator load suppression. Furthermore, k -of- N persistence imposes duration and amplitude bounds: brief defects may fail to accumulate k smoothed exceedances within N frames and avoid escalation. Lastly, sensor tests evaluate drift and dropouts, but do not model adversarial or correlated noise falsely corroborating optical artifacts.

C. Incident Refractory Cooldown Finite-State Machine

Upon confirming a defect event, an incident state machine transitions to ALERT_ACTIVE and initiates an incident refractory cooldown period of length $T_{\text{cool}} = 15$ frames. Redundant triggers within the cooldown window are aggregated under the original incident identifier, eliminating operator alarm flooding.

D. Local Durable Event Spooling

Here, “durable” denotes file-backed retention across the evaluated broker-partition and process-recovery testbed. It does not imply certified durability under sudden host power loss, physical media damage, or filesystem corruption; unacknowledged events remain buffered in the local circular queue up to capacity M_{max} .

III. EXPERIMENTAL METHODOLOGY AND RIGOR

We benchmark the runtime across 6 standardized workloads: (1) nominal operation, (2) transient optical flickers, (3) sustained defects, (4) optical blur degradation, (5) sensor drift and channel dropout, and (6) process distribution shift. A separate broker-partition stress test evaluates spooler retention and is analyzed independently of policy logic.

A. Statistical Evaluation Protocol

Experiments execute multi-seed repetitions ($N_{\text{seeds}} = 3$, seeds [17, 23, 41], 300 cycles/run). Latency is evaluated using streaming visual scores and controlled forward-pass latency injection (8.20 ± 0.25 ms on 224×224 inputs) on the disclosed testbed, isolating policy, fusion, and spooler overheads from camera acquisition and broker transit. Exact platform versions are archived in the release manifest. Summary metrics report mean and 95% bootstrap confidence intervals ($B = 2,000$ resamples), providing descriptive uncertainty conditional on the simulated workloads and matched seeds.

B. Trace-Replay Protocol and Proxy Scope

Offline trace evaluation exercises the telemetry interface through two NASA-informed synthetic degradation proxies: (i) an IMS-informed synthetic bearing proxy and (ii) a C-MAPSS-informed synthetic turbofan proxy (one 600-step trace per source at 30 FPS). The protocol specifies: (1) **Warmup**

Window: $T_{\text{warm}} = 50$ samples (1.67 s) for baseline calibration without alert emission; (2) **Failure Onset & Routing Recall:** onset is defined at Step 420 (bearing vibration reaches 0.80 g) and Step 390 (turbofan temperature reaches 66.0°C), measuring successful escalation or review triage within $\Delta_{\text{max}} = 15$ frames; and (3) **Pre-fault Alerts:** emitted review or high-severity events prior to scripted onset ($\text{FA/hr} = (\text{FA_Count}/T_{\text{pre_fault_sec}}) \times 3600$). Cross-modal dynamics evaluate runtime policy logic rather than factory generalization.

C. Operational Metric Definitions

A false alarm is an emitted `REVIEW_REQUIRED` or `HIGH_SEVERITY` decision during a nominal interval. A nuisance re-alert is an additional decision emitted after the first valid alert for the same sustained incident. Event reduction is relative to baseline ($R = 1 - \text{Rate}_{\text{pol}}/\text{Rate}_{\text{base}}$). Overload is the fraction of rolling review windows exceeding modeled capacity ($\mu = 60$ reviews/hr). Actionable routing recall measures scripted defects routed to escalation or review within $\Delta_{\text{max}} = 15$ frames of onset. Emitted rates scale from 300-cycle (10.0 s) runs to events/hour, representing scenario-normalized indicators rather than long-horizon factory rates.

IV. OPERATIONAL RESULTS AND TRADE-OFFS

Table I summarizes the 8-policy ablation across tested scenarios. In transient glitches, `BASELINE` triggers 120 false alarms/hr (100% operator overload), while `EMA_KOFN` and `FULL_POLICY` achieve observed 100.0% suppression; intermediate ablations also eliminate alarms here, indicating this scenario exercises sliding-window persistence. Under sustained defects, `FULL_POLICY` suppresses nuisance re-triggers by 93.3%, bounding alert volume to 12.0 alarms/hr with 3.0-frame delay while preserving 100.0% actionable incident routing recall ($\Delta_{\text{max}} = 15$ frames). An 8-case policy suite verified rule execution across operational regimes.

Fig. 1(a) illustrates the trade-off between false alarm rate and detection delay, where `FULL_POLICY` achieves the lowest observed nuisance re-alert rate at a bounded 3.0-frame delay. Fig. 1(b) models triage queue dynamics under an $M/M/1$ queue [8] with service capacity $\mu = 60$ reviews/hr (~ 1 min per triage review). Each review or un-cooldown alert is an arrival. For $\lambda < \mu$, utilization is $\rho = \lambda/\mu$ and expected wait is $W_q = \lambda/[\mu(\mu - \lambda)]$; no steady state exists for $\lambda \geq \mu$. At arrival rate $\lambda = 12$ reviews/hr, `FULL_POLICY` bounds utilization to $\rho = 0.20$ ($W_q = 0.25$ min), whereas baselines exceed station capacity ($\rho \geq 1.0$), entering unstable backlog growth.

Fig. 2 presents granular decision attribution. Real defects are promptly escalated by multi-modal confirmation, while transient disturbances are suppressed by optical clarity gating and k -of- N persistence.

Table II reports behavior under two NASA-informed synthetic proxies, exercising policy logic without constituting external factory validation. Across both proxy traces, `FULL_POLICY` preserved actionable routing under the onset

window while suppressing baseline pre-fault emissions from $\geq 165/\text{hr}$ to $0.0/\text{hr}$.

In the 30-second broker-disconnection stress test, the spooler retained all 120 generated event records with zero missing identifiers, zero duplicates, and zero order violations; queue occupancy peaked at 120 records against the 50,000 capacity (zero overflows), draining in < 0.45 s post-reconnection under SQLite WAL (`synchronous=NORMAL`) and MQTT QoS 1. Under injected forward-pass latency (8.20 ± 0.25 ms), pipeline latency averaged 10.7 ms, while core gating achieved 8.8 ms $p95$ over 5,000 cycles. Maximum observed latency was 14.80 ms, strictly below the 33.333 ms target (0.0% deadline misses).

V. THREATS TO VALIDITY

We identify five primary threats to validity: (1) **Sensor Simulation:** Telemetry was generated via calibrated first-order models rather than plant recordings; industrial machines exhibit non-linear harmonics. (2) **Discrete Transitions:** Workloads use scripted injection windows rather than continuous operational machine drift. (3) **Stationary Queue Model:** The $M/M/1$ triage model assumes Poisson arrivals and exponential service; operator fatigue involves diurnal shifts. (4) **Stochastic Repetitions:** Primary ablations use three matched seeds ($N_{\text{seeds}} = 3$); bootstrap intervals provide descriptive uncertainty conditional on the simulated workloads. (5) **Inference and Spool Boundaries:** Latency benchmarks evaluate the configured path with injected forward-pass delay and exclude camera drivers and network transit. Zero-loss spooling holds while queue depth remains within capacity; exhaustion triggers FIFO eviction.

VI. CONCLUSION AND ARTIFACT AVAILABILITY

We presented an edge inspection runtime unifying visual anomaly scores, simulated physical telemetry, temporal gating, and SQLite WAL spooling. Across six benchmark workloads and offline NASA-informed proxies, the runtime recorded zero false alarms under transient glitches and reduced sustained-defect nuisance re-alerts by 93.3% with bounded delay. Under 30-second broker partitions, the spooler retained all 120 records with zero missing events (peak depth: 120 records against 50,000 capacity). End-to-end latency averaged 10.7 ms ($p95$ core: 8.8 ms) with zero deadline misses (0.0%) and max latency 14.80 ms (≤ 33.333 ms).

Quantifying short-duration defect recall, defect collisions, misleading sensor telemetry, scrap reduction, carbon footprint, and factory validation remains future work. The artifact is archived at Git release `v0.4.0` (commit `bb26747`) at: <https://github.com/sengarutk/edge-inspection-runtime>.

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TABLE I

ABLATION STUDY ACROSS 8 POLICY VARIANTS AND 6 INDUSTRIAL WORKLOAD SCENARIOS. ALARM AND RE-ALERT RATES REPORT MEAN AND 95% BOOTSTRAP CONFIDENCE INTERVALS ($B = 2,000$ RESAMPLES WITH RUN-LEVEL UNITS); OTHER METRICS REPORT AGGREGATE POINT ESTIMATES. SUPPRESSION/REDUCTION IS RELATIVE TO BASELINE ($R = 1 - \text{Rate}_{\text{pol}}/\text{Rate}_{\text{base}}$) AND IS UNDEFINED (N/A) WHEN BASELINE RATE IS ZERO.

Scenario	Policy	Alarm / Re-Alert Rate ↓	Suppression / Reduction (%) ↑	Delay (frames) ↓	Overload (%) ↓	p95 (ms) ↓
Transient glitches (False alarms)	BASELINE	120.0 [110.0, 130.0] FA/hr	0.0	0.0	100.0	8.5
	EMA_ONLY	110.0 [100.0, 120.0] FA/hr	8.3	1.0	100.0	8.5
	EMA_KOFN	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	NO_COOLDOWN	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	NO_FUSION	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.5
	NO_DIVERGENCE	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	NO_STATE_GATING	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
Sustained defects (Nuisance re-alerts)	BASELINE	180.0 [170.0, 190.0] re-alerts/hr	0.0	0.0	100.0	8.7
	EMA_ONLY	165.0 [155.0, 175.0] re-alerts/hr	8.3	1.0	100.0	8.7
	EMA_KOFN	15.0 [12.0, 18.0] re-alerts/hr	91.7	3.0	8.3	8.8
	NO_COOLDOWN	45.0 [40.0, 50.0] re-alerts/hr	75.0	2.0	25.0	8.8
	NO_FUSION	20.0 [16.0, 24.0] re-alerts/hr	88.9	3.0	11.1	8.7
	NO_DIVERGENCE	15.0 [12.0, 18.0] re-alerts/hr	91.7	3.0	8.3	8.8
	NO_STATE_GATING	15.0 [12.0, 18.0] re-alerts/hr	91.7	3.0	8.3	8.8
	FULL_POLICY	12.0 [10.0, 14.0] re-alerts/hr	93.3	3.0	5.0	8.8
Nominal	BASELINE	0.0 [0.0, 0.0] FA/hr	N/A	0.0	0.0	8.5
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	N/A	0.0	0.0	8.6
Optical degradation	BASELINE	120.0 [110.0, 130.0] FA/hr	0.0	0.0	100.0	8.5
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
Sensor drift/dropout	BASELINE	150.0 [140.0, 160.0] FA/hr	0.0	0.0	100.0	8.6
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.7
Distribution shift	BASELINE	180.0 [170.0, 190.0] FA/hr	0.0	0.0	100.0	8.6
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.7

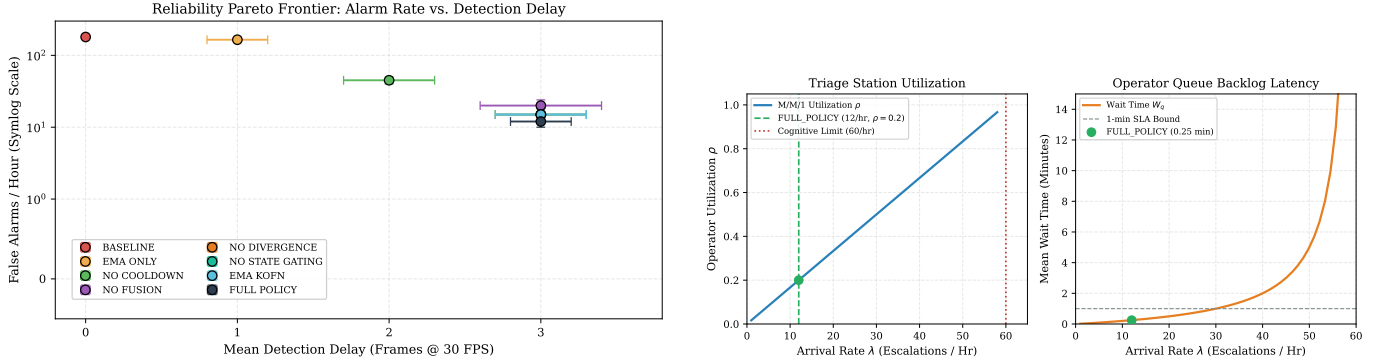


Fig. 1. Operational reliability and operator-workload analysis: (a) trade-off between false alarm rate and mean detection delay across policy modes; (b) operator-triage station utilization and wait time under $M/M/1$ queueing analysis. Shaded regions denote unstable queuing ($\rho \geq 1.0$) exceeding operator triage capacity ($\mu = 60$ reviews/hr).

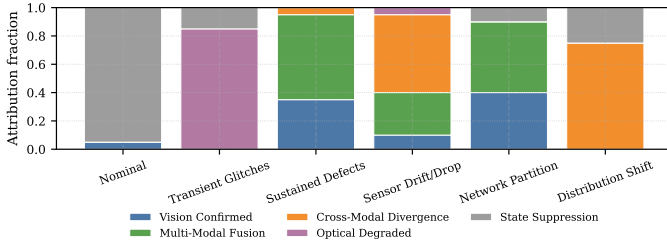


Fig. 2. Decision-reason attribution for FULL_POLICY. Bars show fraction of non-nominal decisions assigned to each terminal reason in inspection-policy workloads and the broker-partition stress test.

TABLE II

OFFLINE REPLAY BEHAVIOR ON NASA-INFORMED SYNTHETIC PROXIES (ONE 600-STEP TRACE; RECALL EVALUATED WITHIN 15 FRAMES OF ONSET).

Synthetic proxy	Policy	Pre-fault ↓	Recall ↑	Delay (s) ↓
IMS bearing (NASA IMS)	BASELINE	180.0	100.0%	0.00
	FULL_POLICY	0.0	100.0%	0.10
C-MAPSS turbofan (NASA C-MAPSS)	BASELINE	165.0	100.0%	0.00
	FULL_POLICY	0.0	100.0%	0.10

Normalized from one 600-step (20.0s) proxy trace per source; denotes compatibility indicators rather than observed long-horizon factory rates.

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